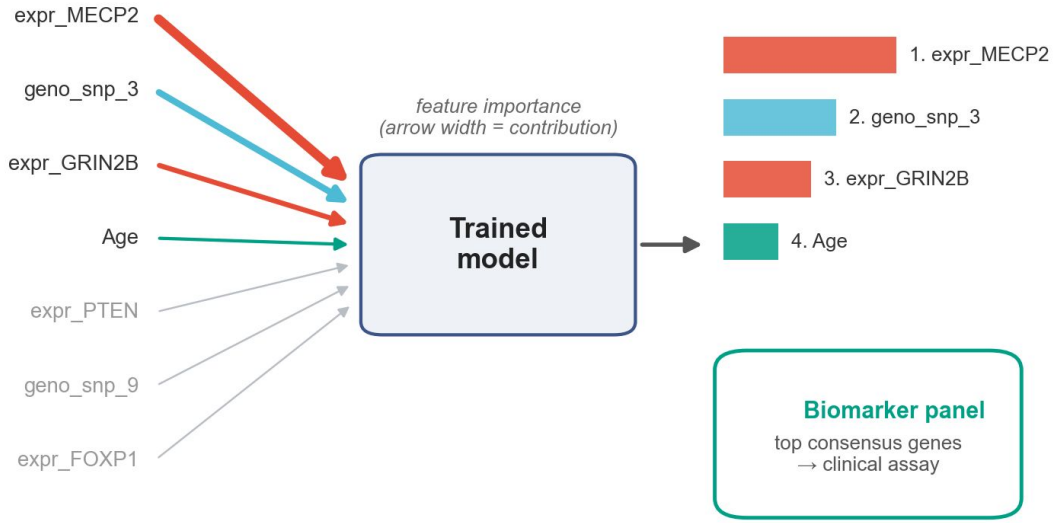
Lesson 7: Explainable AI & Feature Importance

From trained models to ranked features, explanations,

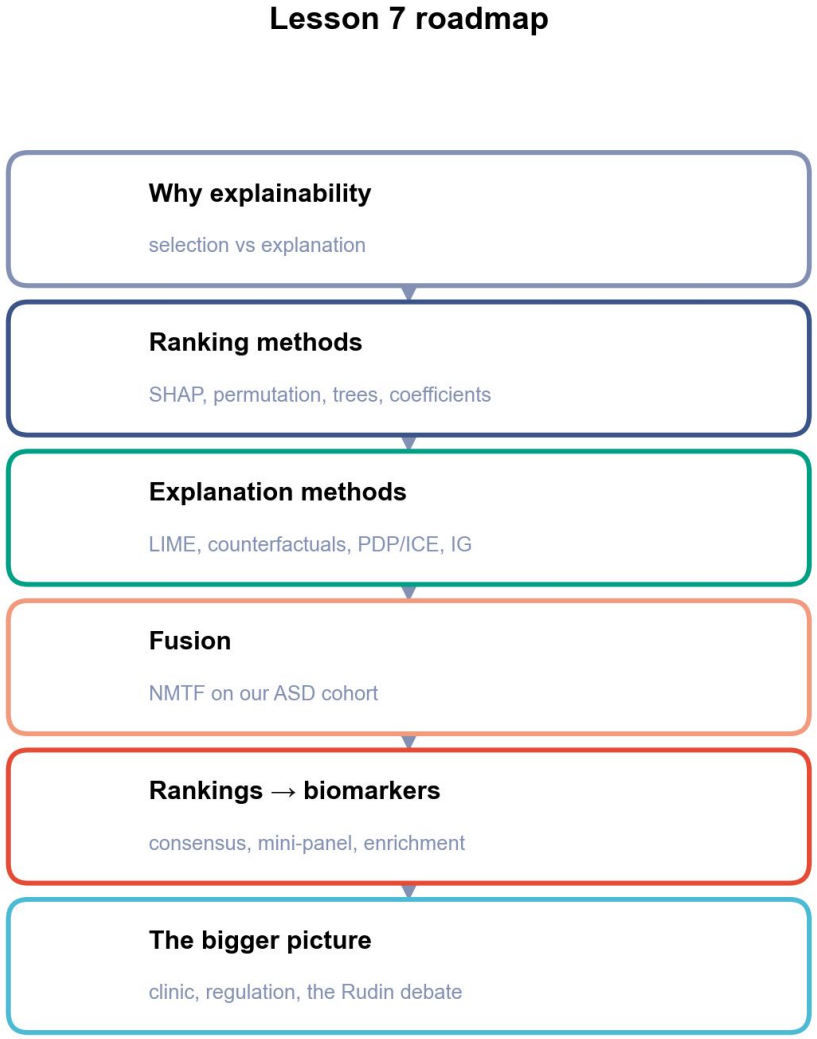
- and validated biomarker panels

Nine methods, one cohort: everything runs on our ASD dataset



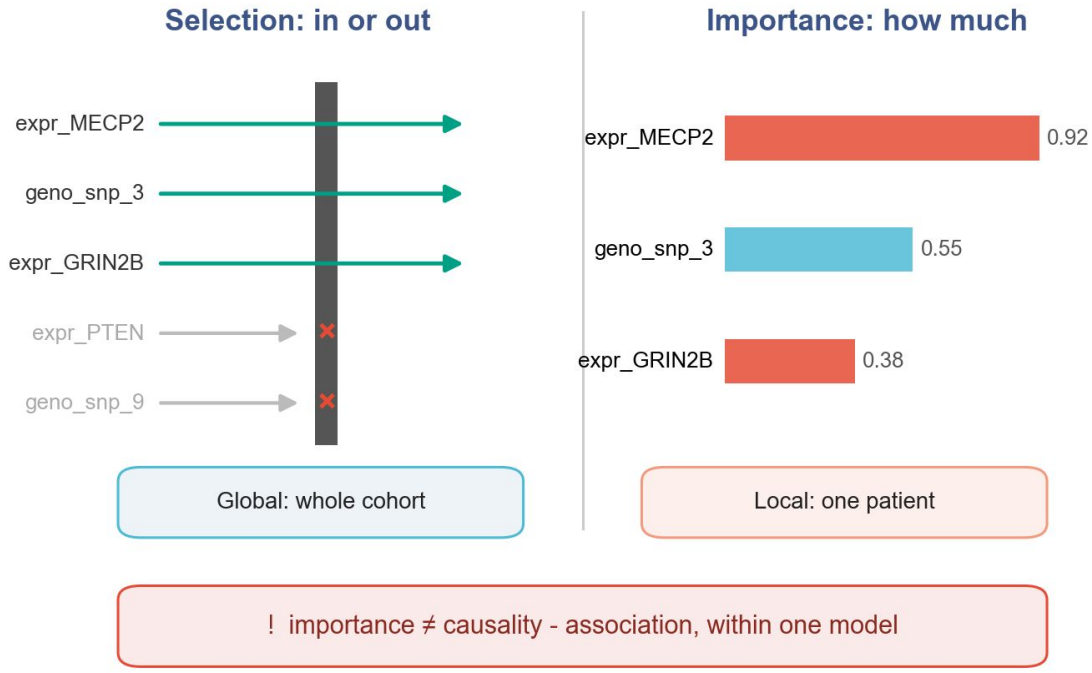
Agenda: The Road from Model to Biomarker

- 1 | Why explainability - and how it differs from feature selection
- 2 | Ranking methods: SHAP, permutation, tree importances, coefficients
- 3 | Explanation methods: LIME, counterfactuals, PDP/ICE, integrated gradients
- 4 | Fusion: NMTF tri-factorization, applied to our ASD cohort
- 5 | Rankings → biomarkers: consensus, mini-panel, pathway enrichment
- 6 | The bigger picture: clinical translation, regulation, the Rudin debate



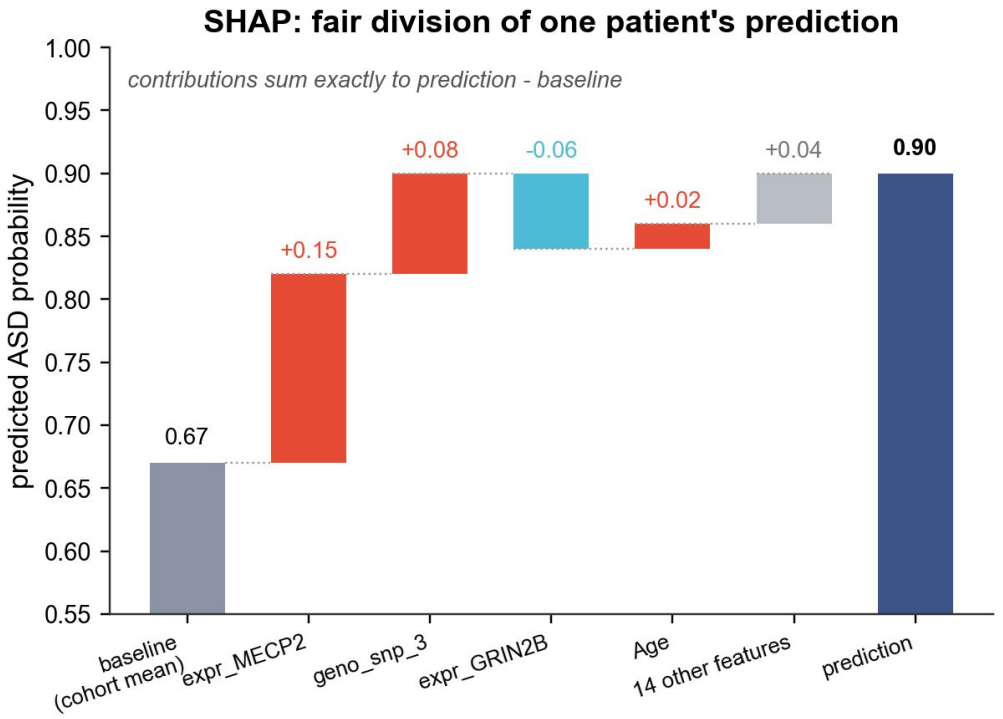
How This Differs from Lesson 6: Selection vs Explanation

- Lesson 6 (selection): binary in/out BEFORE or DURING training
  - goal: shrink p, fight overfitting; output: a feature list
- Lesson 7 (importance/XAI): interrogate a TRAINED model
  - goal: rank, understand, justify; output: scores + explanations
- New axis today: global (whole cohort) vs local (THIS patient)
- Warning carried over: importance ≠ causality (association, in-model)



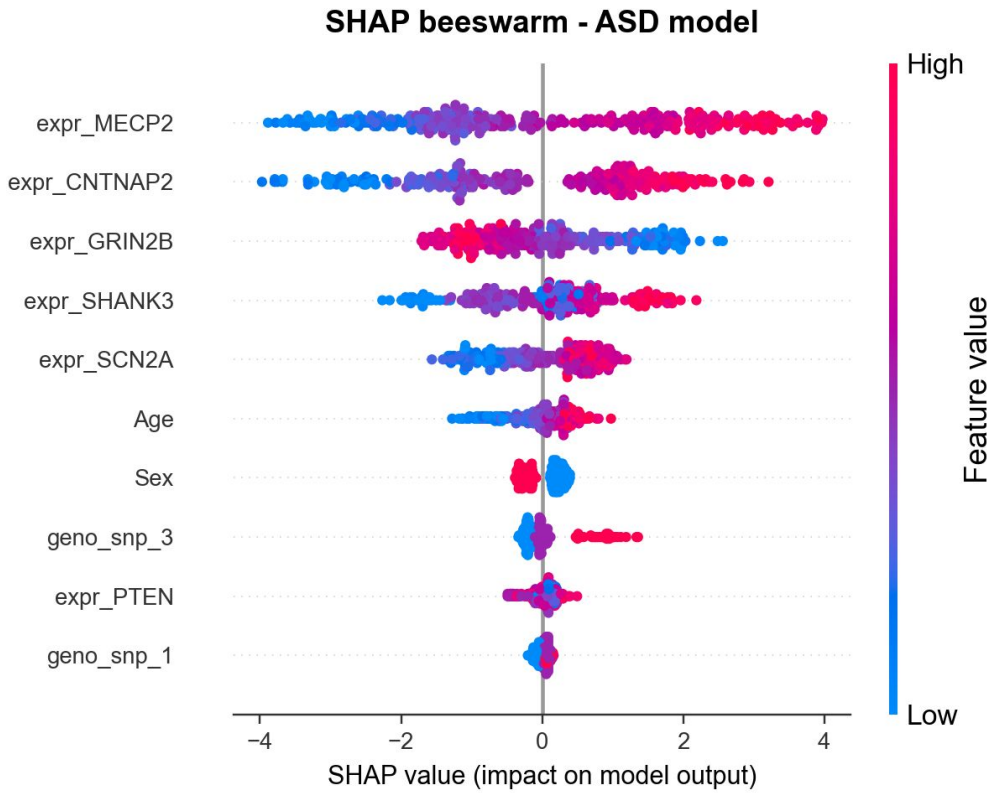
Method 1 - SHAP Values: The Theory

- SHapley Additive exPlanations: fair payout division from cooperative game theory
- Players = features, payout = this sample's prediction
- SHAP value = feature's fair contribution to THIS prediction
- Guarantee: contributions sum exactly to prediction - baseline
- Local (per-patient) AND global (averaged) in one framework
- Paper: Lundberg & Lee 2017, NeurIPS - ~35,000 citations
- Game theory finally explains something: your classifier



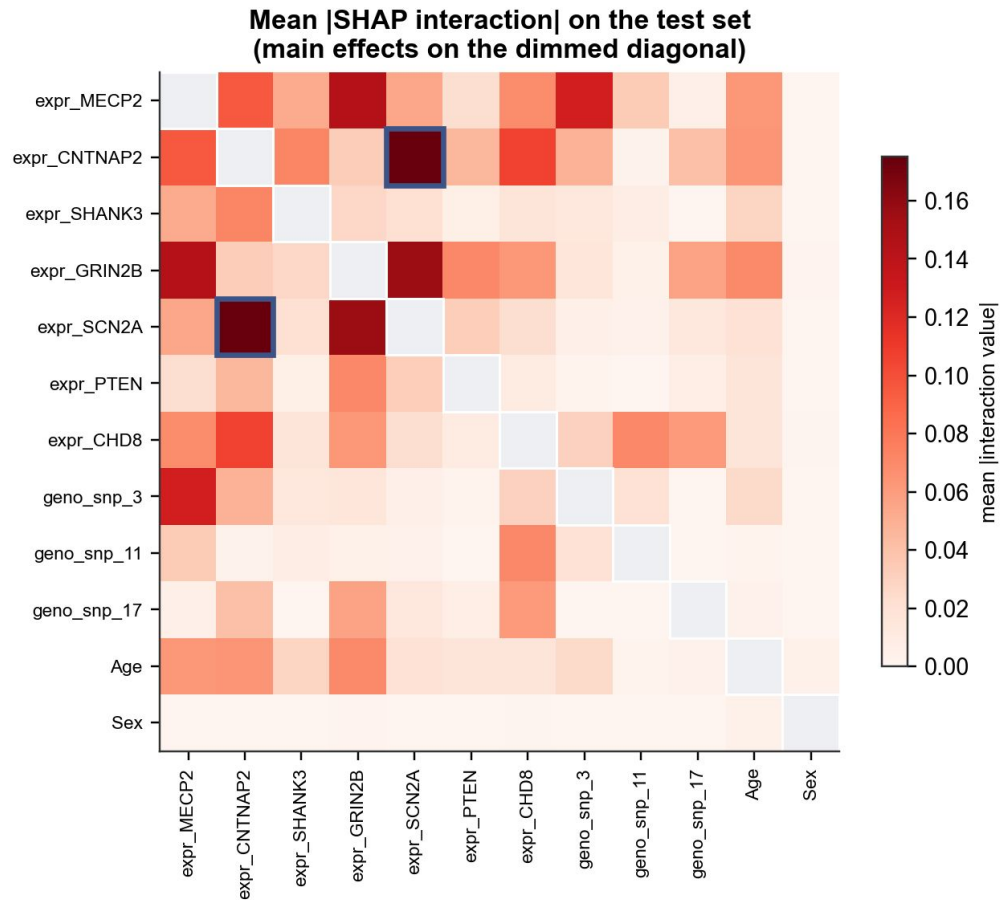
SHAP in Practice: The Plots

- Beeswarm: global ranking + direction + per-patient spread
    - each dot = one patient; color = feature value; x = SHAP
- Dependence plot: SHAP vs feature value → shape & interactions
- Waterfall/force: single-patient explanation for the clinic
- Read: 'high MECP2 (red, right) pushes toward ASD'
- How to read beeswarm plot?
    - Clear Color Separation: If all red dots are on the right and all blue dots are on the left, the feature has a straightforward positive correlation with the target
    - Long Tails: A few dots stretching far to the left or right indicate that for specific edge cases, that feature has an extreme impact on the prediction, even if it has little effect on average.
    - Mixed Colors: If red and blue dots are scattered together, it usually indicates a non-linear relationship or strong interaction effects with other features in the model.



SHAP Interaction Values: Epistasis Inside the Model

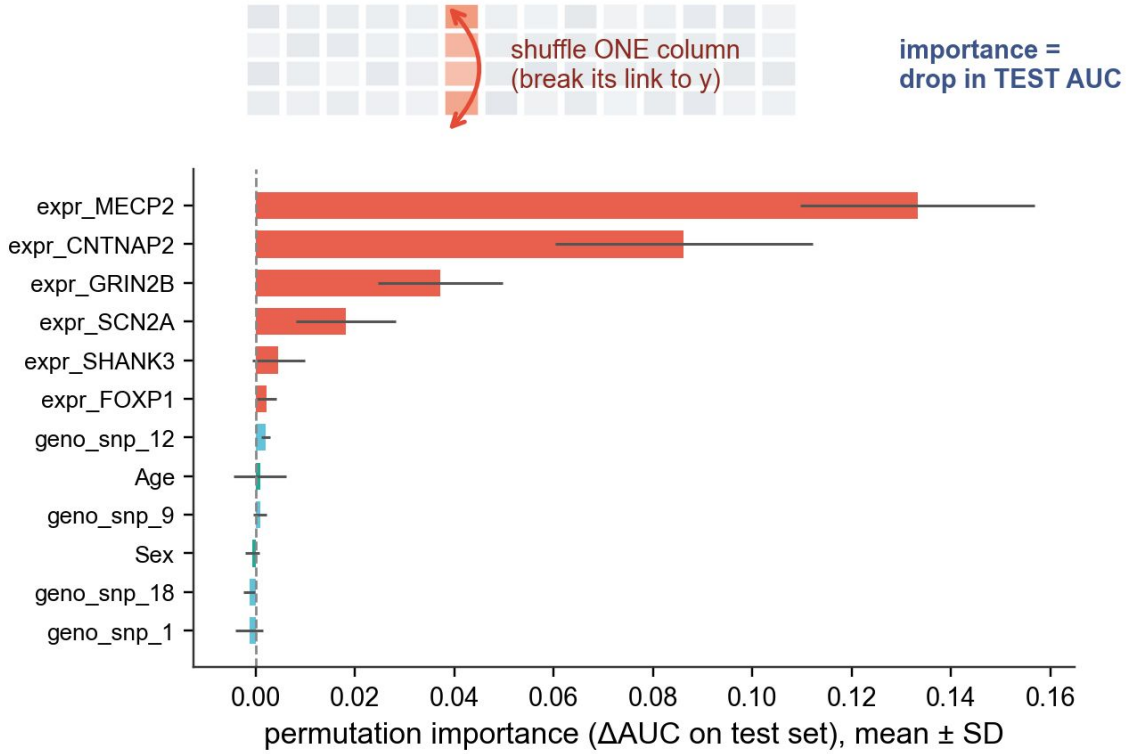
- SHAP splits each prediction into per-feature main effects...
- ...interaction values split it further into per-PAIR effects
- Exact for trees: explainer.shap_interaction_values(X)
- Heatmap of mean |interaction| = the model's epistasis map
- A SNP x expression interaction = candidate regulatory mechanism
- Paper: Lundberg et al. 2020, Nature Machine Intelligence
- Cost: p x p matrix per patient - use on panels, not transcriptomes



outlined: strongest off-diagonal pair = expr_CNTNAP2 x expr_SCN2A

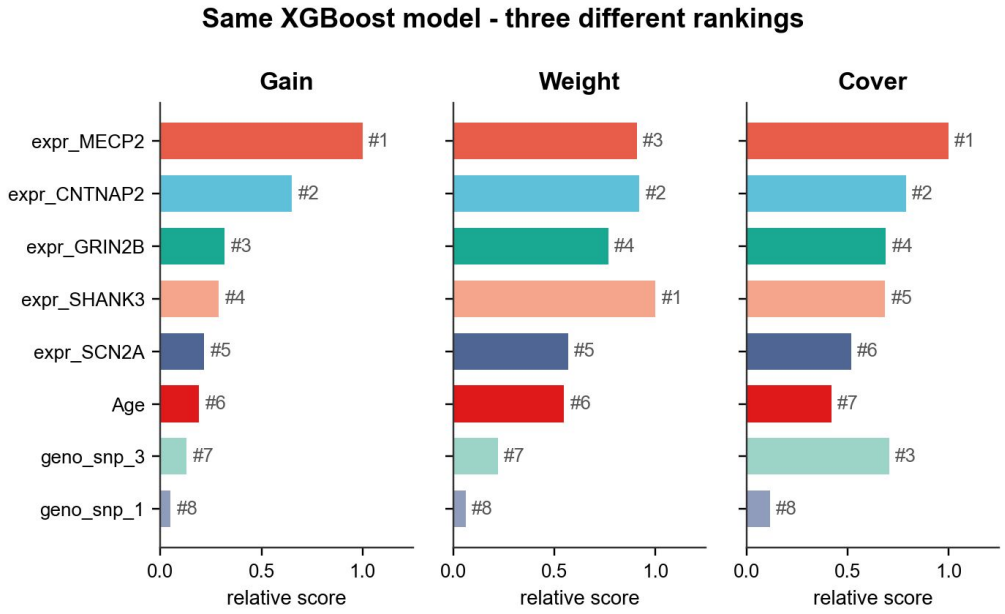
Method 2 - Permutation Importance (Gold Standard)

- Shuffle one feature's column → break its link to the outcome
- Importance = drop in TEST performance after shuffling
- Model-agnostic: works for any fitted model, any metric
- Computed on held-out data → no training-data optimism
- Repeat shuffles → mean ± SD, honest error bars
- Paper: Breiman 2001, 'Random Forests' - ~130,000 citations
- Shuffle a gene, watch the model panic



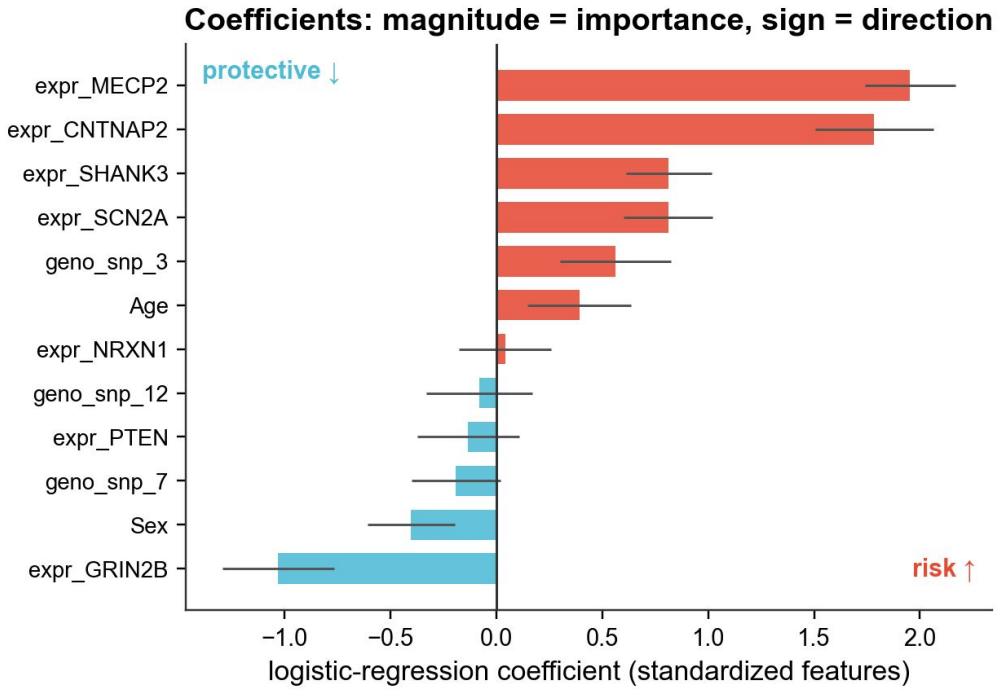
Method 3 - Tree-Based Importances: Gini, Gain, Weight, Cover

- Trees rank features as a by-product of training - free and instant
- Gini importance (RF): total impurity reduction per feature
- Boosting flavors: Gain (loss improvement - the meaningful one),
  - Weight (split counts), Cover (samples touched) - rankings differ!
- Biases: favors high-cardinality features; computed on TRAINING data
- Papers: CART book 1984 - ~60,000; XGBoost 2016 - ~40,000 citations
- Use for triage; always state WHICH flavor; confirm with Method 2



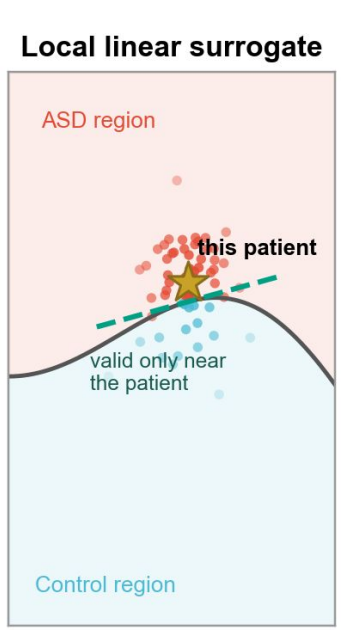
Method 4 - Linear Model Coefficients

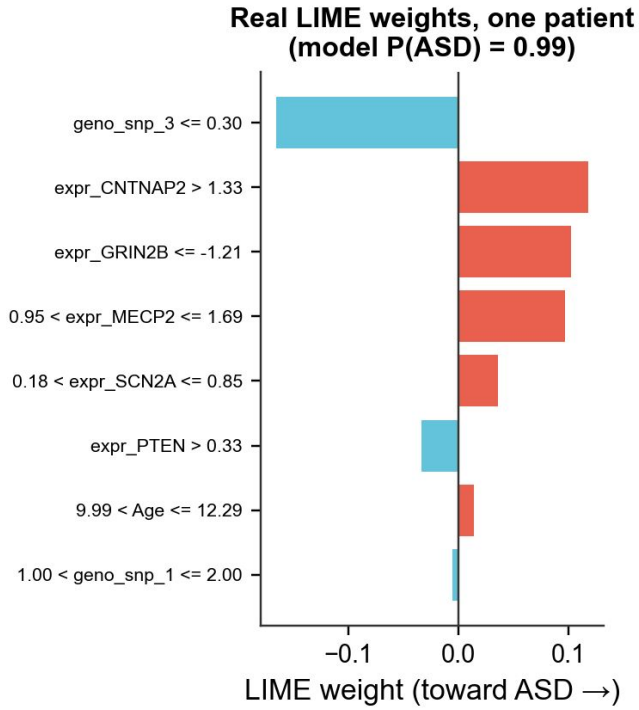
- Standardized features → |coefficient| = importance
- Sign gives DIRECTION: risk ↑ or protective ↓
- LogReg: coefficient = change in log-odds per +1 SD of feature
- Most transparent method - clinicians can audit it
- Requires standardization, else units corrupt the ranking
- No single founding paper - classical statistics (Cox 1958)
- Correlated genes destabilize coefficients (use ElasticNet)



Method 5 - LIME: Local Surrogate Models

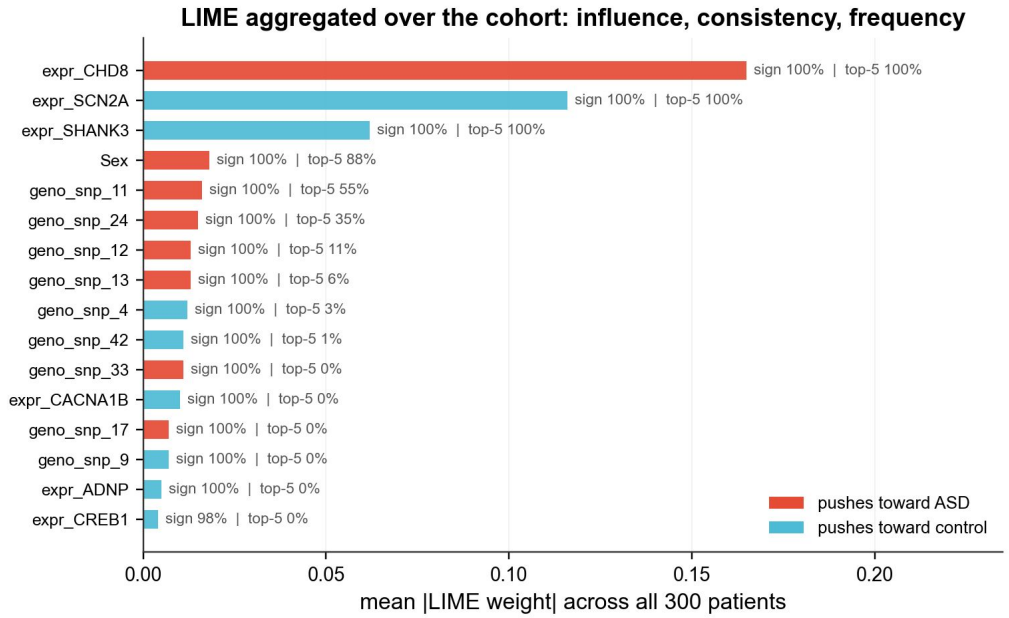
- LIME = Local Interpretable Model-agnostic Explanations
- Perturb one patient's features → query the black box →
  - fit a simple LINEAR model valid only in that neighborhood
- Output: per-patient weights - 'why THIS patient?'
- vs SHAP: faster, intuitive; but no additivity guarantee, unstable
- Paper: Ribeiro et al. 2016, KDD - ~25,000 citations
- Use for quick clinical case review; SHAP for publication





LIME at Cohort Scale: Consistency, Frequency, Global View

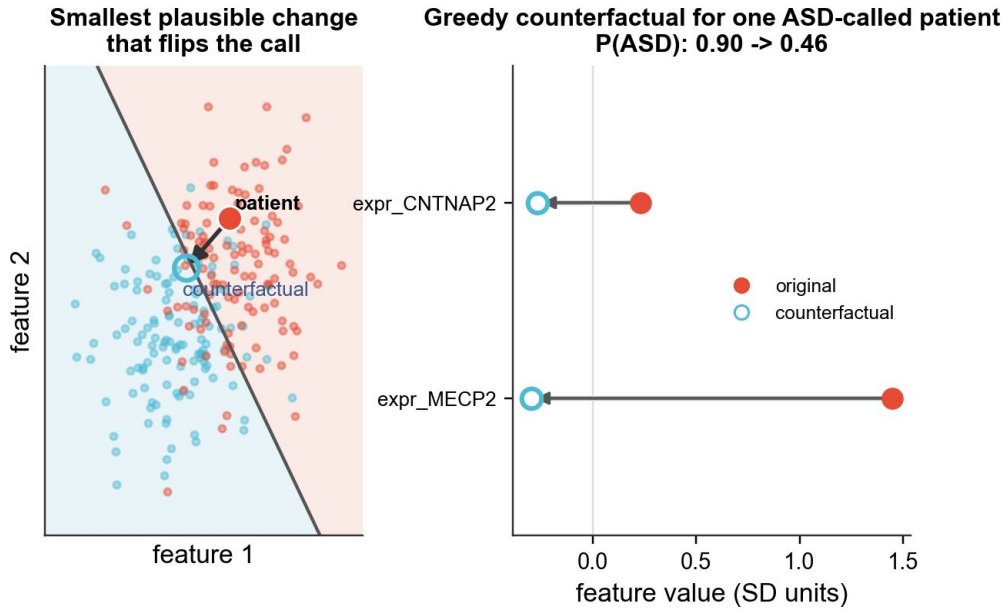
- One patient's LIME story is an anecdote - so run ALL 300 patients
- Score each attribute cohort-wide: influence (mean |w|),
  - sign consistency, and top-5 frequency (enters the short story?)
- CHD8, SCN2A, SHANK3 in EVERY patient's top-5; Sex 88%, snp_11 55%
- Sign consistency ~100%: each attribute pushes ONE way, cohort-wide
- vs permutation (same RF): rho = 0.51 - top agrees, noise tail scrambles
- SP-LIME idea (Ribeiro 2016); 300 x 5,000 queries - own the trees? TreeSHAP



agreement with permutation importance on the same RF: Spearman rho = 0.51 (top 3 agree; the near-zero tail scrambles)

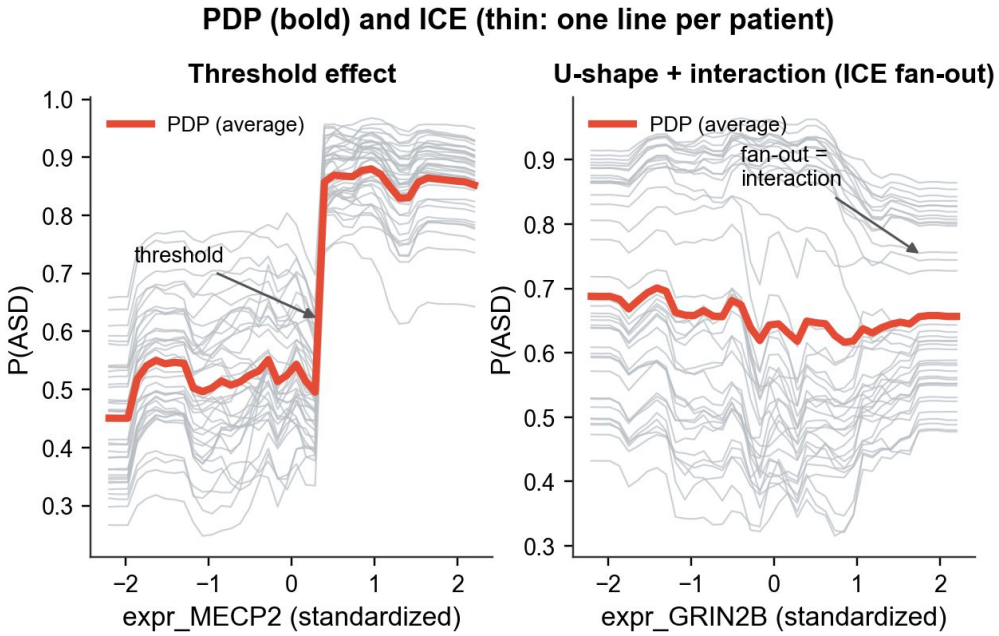
Method 6 - Counterfactuals: What Would Flip the Call?

- Explanation as a recipe: smallest change that flips the prediction
- 'If CHD8 expression were 1 SD lower, the model would say control'
- The native language of clinical reasoning - no math required to read
- Search: nudge features (importance-ordered) until the label flips
- Must respect plausibility: no negative ages, no impossible dosages
- Paper: Wachter et al. 2017, Harvard JOLT - ~7,000 citations
- Caveat inherited from Lesson 6: model recipe ≠ biological intervention



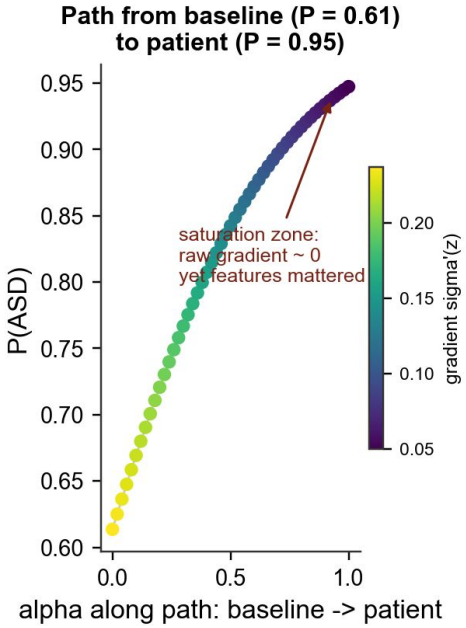
Method 7 - Partial Dependence & ICE Plots

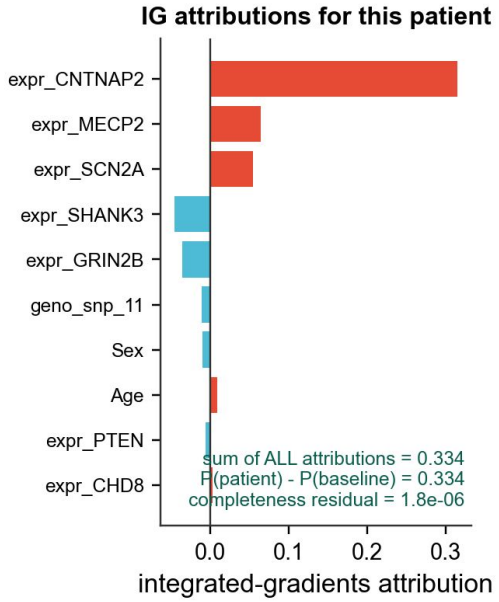
- PDP: average prediction as ONE feature sweeps its range
    - (all other features marginalized out)
- ICE (Individual Conditional Expectation): one thin line per patient - the individual version
- Reveals effect SHAPE: linear, threshold, U-shape, saturation
- ICE lines fanning out = interaction with other features
- Papers: Friedman 2001 - ~40,000; ICE: Goldstein 2015 - ~2,500
- Caveat: assumes feature independence - genes are NOT independent



Method 8 - Integrated Gradients: Attribution for Neural Nets

- The gradient question: how does the prediction change per feature?
- Raw gradients saturate; IG integrates them along a path
  - from a BASELINE (e.g. cohort mean) to the patient
- Completeness: attributions sum exactly to prediction - baseline
- Standard for deep omics models (expression nets, DNA sequence CNNs)
- Paper: Sundararajan et al. 2017, ICML - ~10,000 citations
- For trees/linear you have SHAP; IG is your tool when networks arrive





Method 9 - NMTF: Tri-Factorization for Multi-Omics Fusion

• Messa et al. 2024 (Int J Mol Sci, CC BY): representation learning

- by Non-negative Matrix Tri-Factorization of relation matrices - unsupervised!

• Each matrix: **R ≈ U x S x V^T**

- U, S, V low-rank and non-negative

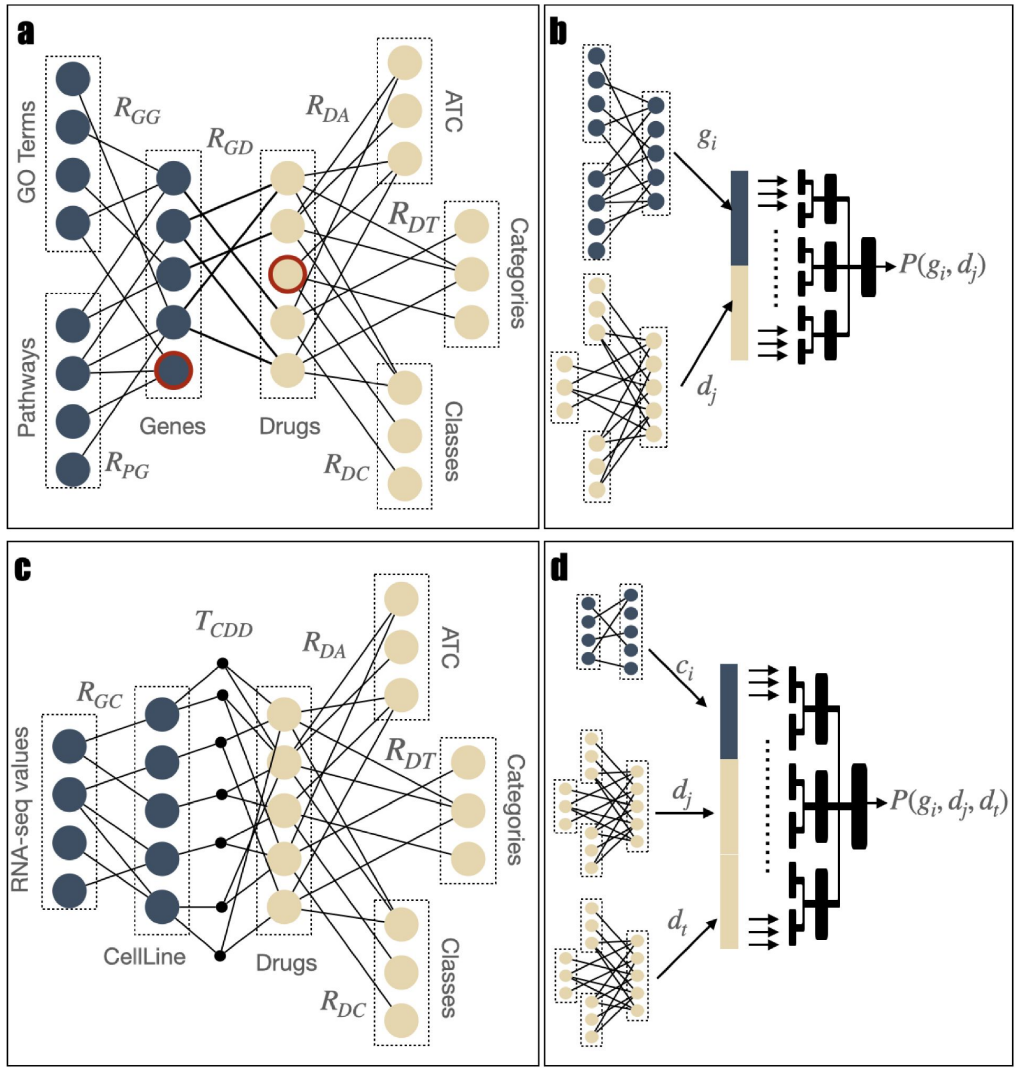
• Fusion trick: node sets SHARED by matrices share ONE factor matrix

• Lineage: tri-factorization Ding et al. 2006, KDD - ~2,000 citations

• Not post-hoc XAI: interpretable BY DESIGN - factors ARE the explanation

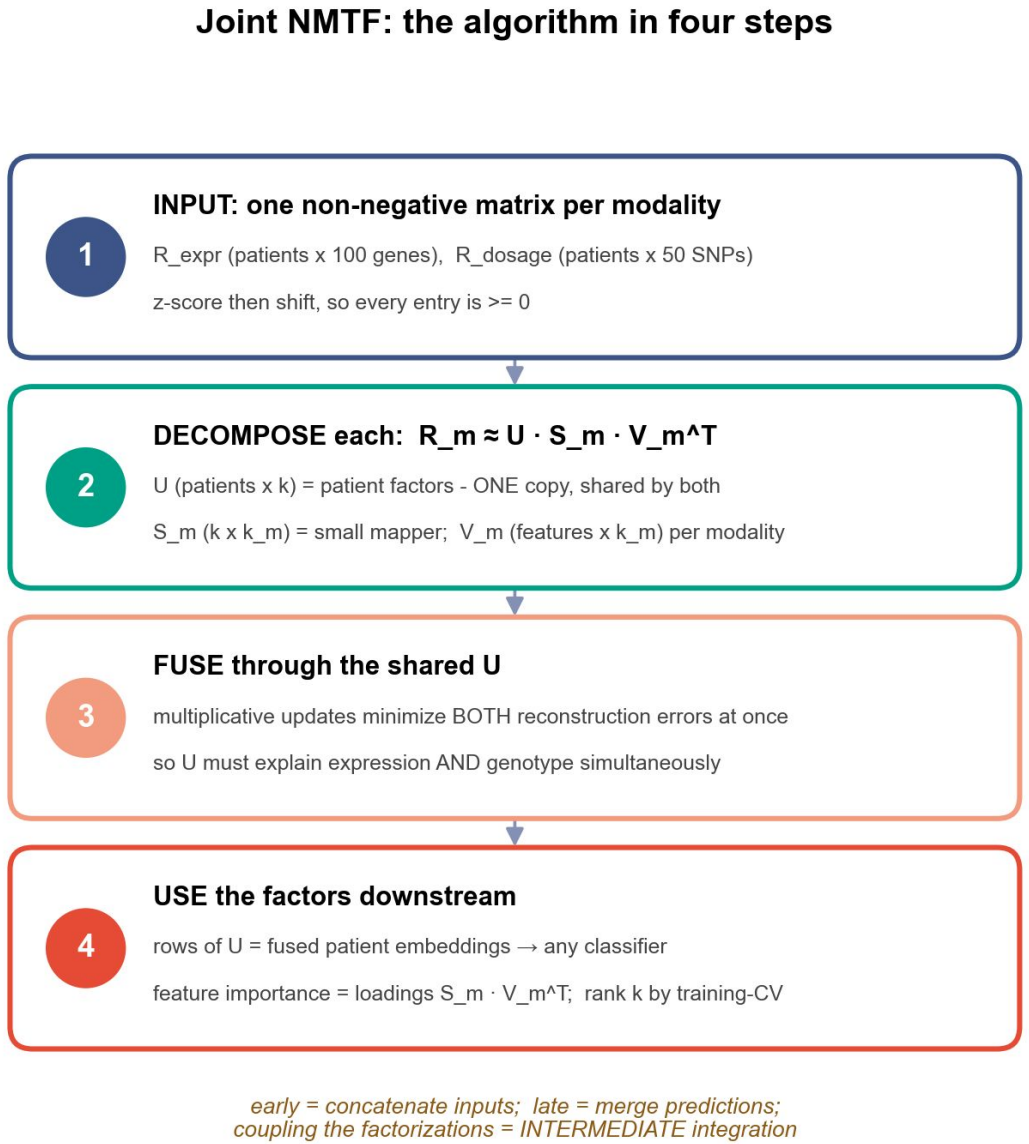
• Figure: their drug-repurposing multipartite graphs (a, c) and

- the embedding + classifier pipeline (b, d)



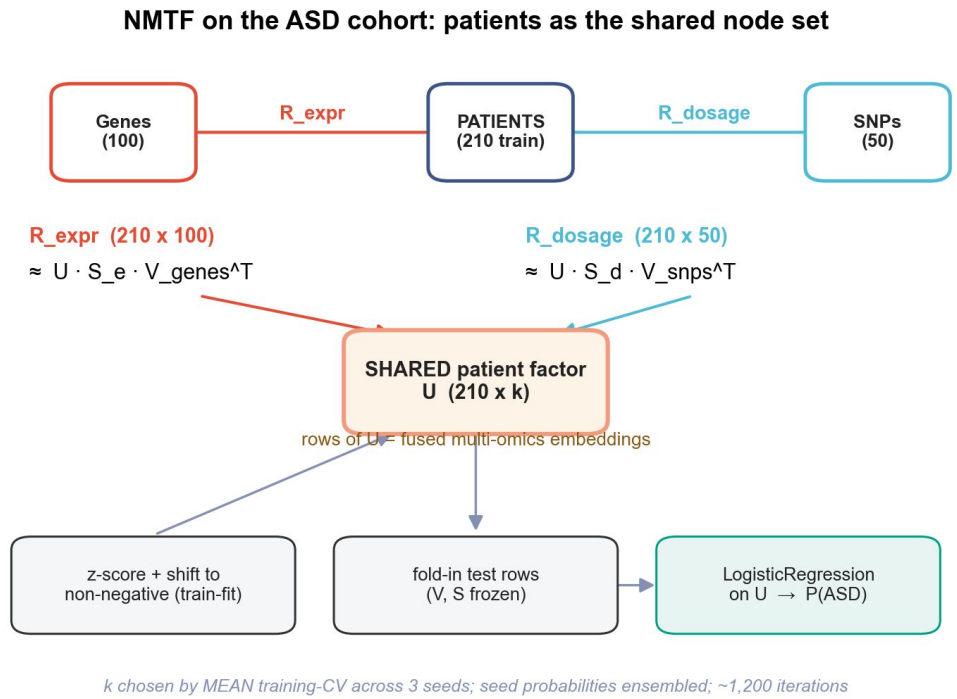
NMTF Under the Hood: The Algorithm in Four Steps

- 1 INPUT: one non-negative matrix per modality (patients x features)
- 2 DECOMPOSE each: R_m ≈ U · S_m · V_m^T - with ONE shared U
  - U = patient factors; S_m = small mapper; V_m = gene / SNP factors
- 3 FUSE: multiplicative updates minimize BOTH errors at once - the SUM of the two Frobenius reconstruction errors
  - so U must explain expression AND genotype simultaneously
- 4 USE: rows of U = embeddings → classifier; loadings - feature importance - read off the reconstructed loadings S_m x V_m^T



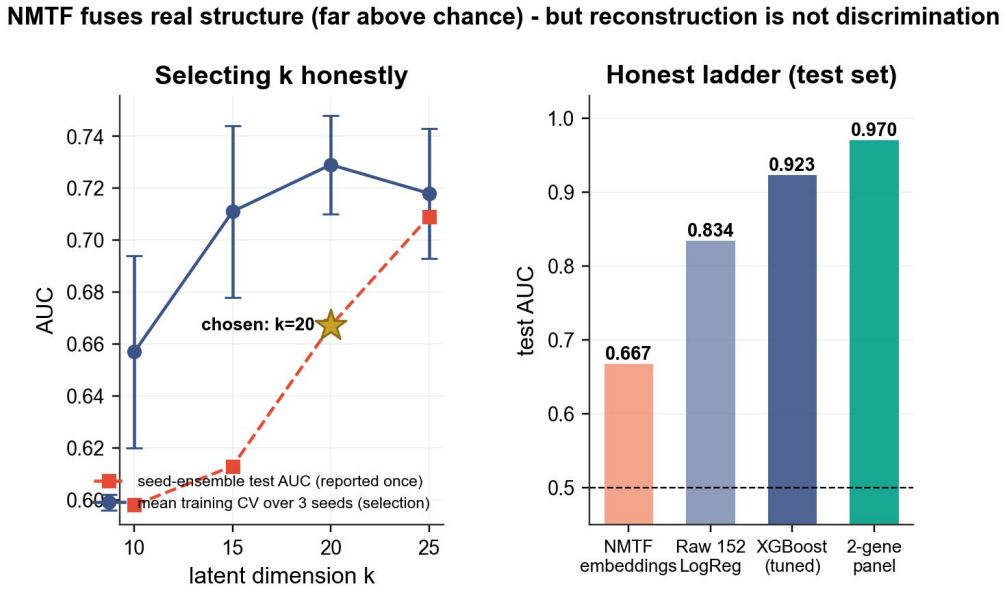
NMTF on Our ASD Cohort: How We Applied It

- Our tripartite graph: Genes - PATIENTS - SNPs; patients are shared
- R_expr (210 x 100) ≈ U S_e V_genes^T; R_dosage (210 x 50) ≈ U S_d V_snps^T
    - with ONE shared patient factor U = the fused representation
- Non-negativity via z-score-then-shift; ~30 lines of NumPy updates
- Test patients: FOLD-IN (V, S frozen; only new U rows fitted) - no leakage
- k by MEAN training-CV across 3 seeds; seed probabilities ensembled



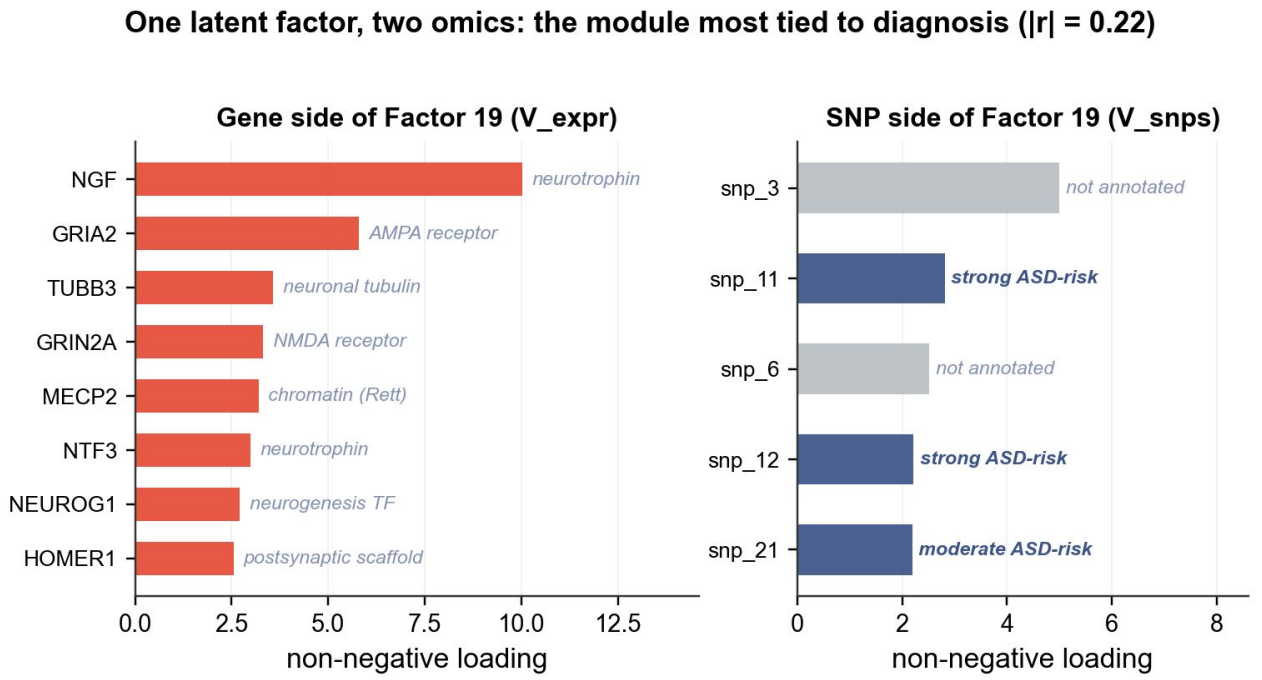
NMTF on ASD: Results - It Fuses Structure, Not Labels

- Mean train-CV over 3 seeds picks k = 20 → ensembled test AUC 0.67
- Honest ladder: NMTF 0.67 < raw LogReg 0.83 < XGBoost 0.92 < panel 0.97
- Why: factors maximize RECONSTRUCTION; the ASD signal lives in ~5 of
  - 152 features - too little variance to earn a factor of its own
- What NMTF DID deliver: interpretable non-negative gene/SNP modules
- Our ASD signal lives in a handful of genes among 152 features; it simply does not carry enough variance to earn its own factor, so the label-relevant information gets diluted across components -



NMTF as XAI: A Cross-Omics Module You Can Read

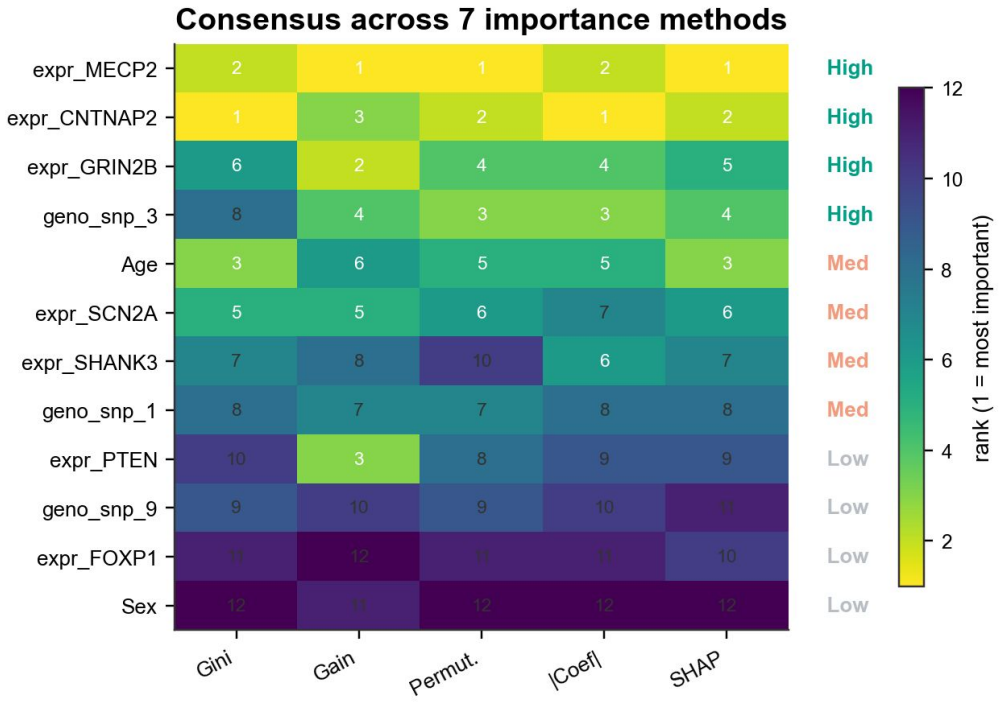
- Factor most tied to diagnosis: #19 (|r| = 0.22, training set)
- Gene side reads as biology: NGF, NTF3 (neurotrophins); GRIA2
- R_m ≈ U S_m V_m^T: every value = an additive sum of modules;
  - a patient's row of U = their personal module mix
- Caveat: modules capture VARIANCE, not the label (r only 0.22)



non-negative loadings are parts that ADD UP: the factorization itself is the explanation (SNP risk labels: ASD_snp_annotations.csv)

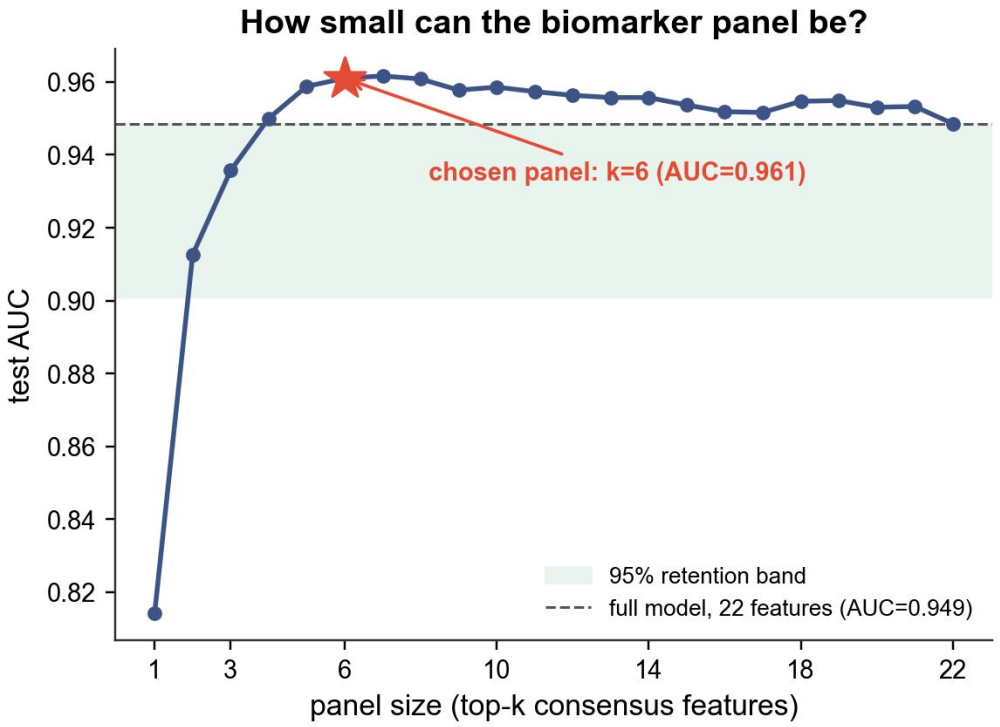
Consensus: Aggregating the Ranking Methods

- No method is unbiased - but their biases differ
- Rank features under every method → average / vote
- Consensus tiers: all methods (high), most (medium), one (low)
- Bootstrap on top: does the gene survive resampling too?
- Stable consensus features = biomarker candidates
- Divergences are diagnostics, not noise (redundancy, interactions)



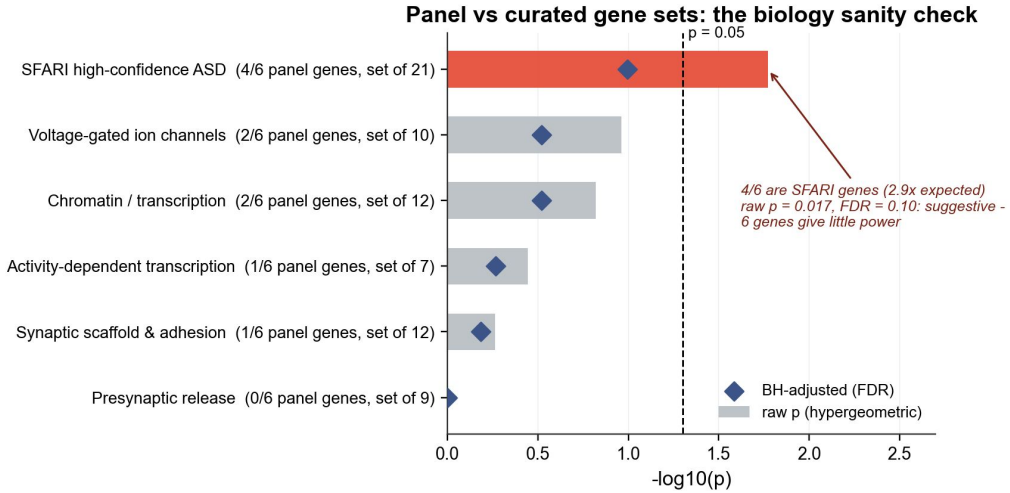
From Importance to a Validated Mini-Panel

- Step 1: consensus top 10-20 features → candidate panel
- Step 2: retrain using ONLY the panel
- Step 3: compare with full model on untouched test data
- Keep if panel retains ≳95% of full performance
- ASD notebook: 16 features, AUC 0.95 → 2-gene panel, AUC 0.97
- 20,000 genes auditioned, 15 made the band - one plays lead guitar



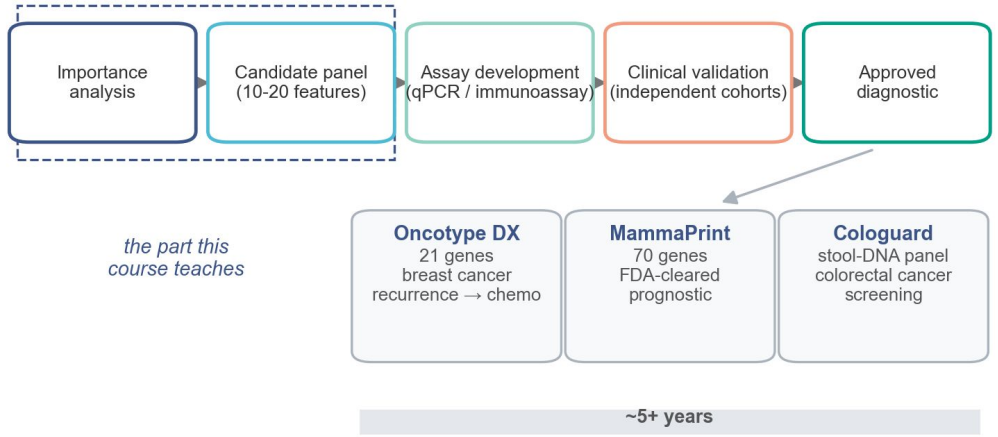
From Panel to Biology: Pathway Enrichment

- The explanation a biologist trusts: does the panel make SENSE?
- Hypergeometric test: panel genes vs curated gene sets (the GSEA idea)
- Our panel's 6 genes: 4 are SFARI ASD genes (2.9x expected), raw p 0.017
    - but BH FDR 0.10 across 6 sets - tiny panels have little power!
- Papers: GSEA - Subramanian et al. 2005, PNAS - ~40,000 citations;
    - web tools: Enrichr (2016), g:Profiler (2019)
- XAI output = hypothesis; enrichment = the biological sanity check



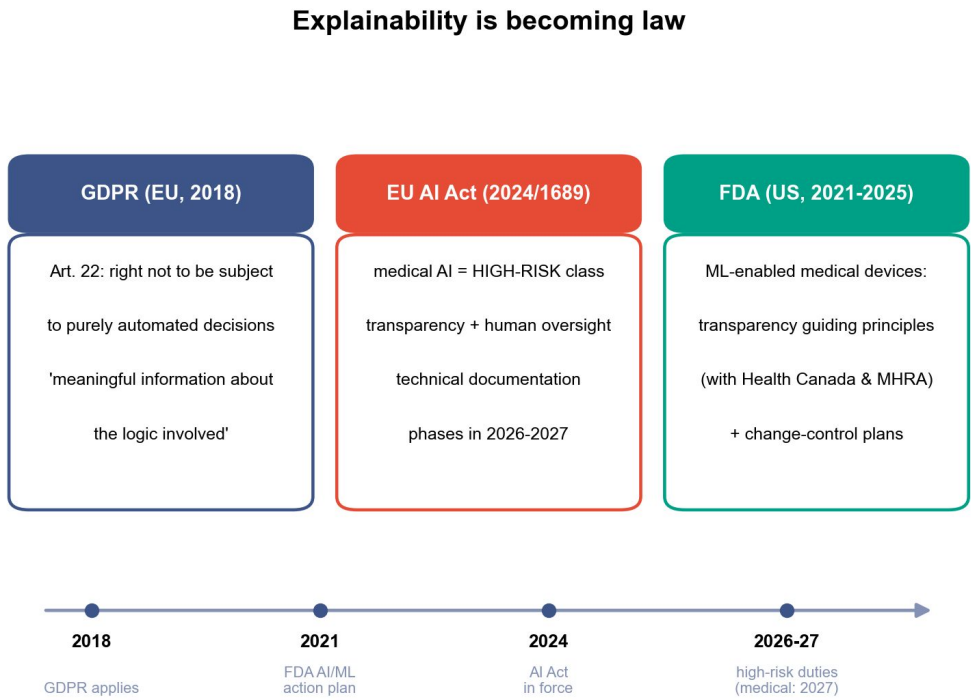
Clinical Translation: Real-World Panels

- Oncotype DX: 21 genes - breast cancer recurrence, guides chemo
- MammaPrint: 70 genes - FDA-cleared prognostic signature
- Cologuard: stool DNA panel - colorectal cancer screening
- Path: importance → panel → assay (qPCR) → independent cohorts
    - → analytical + clinical validation → regulatory approval
- Every one of them started life as a feature-importance table



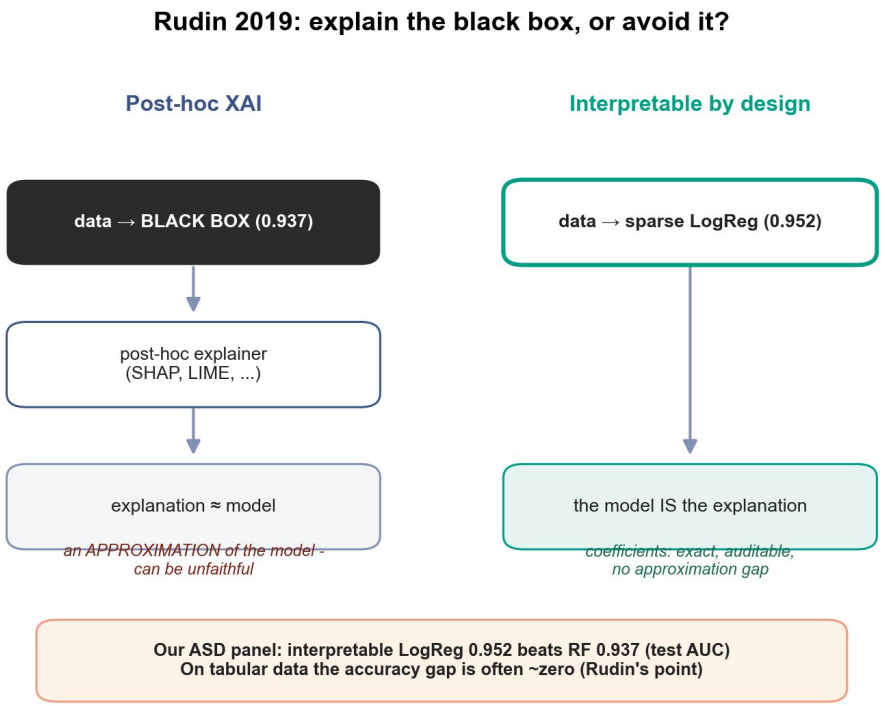
Explainability Is Becoming Law

- GDPR (EU, 2018), Art. 22: limits purely automated decisions;
  - 'meaningful information about the logic involved' (Wachter's trigger)
- EU AI Act (Reg. 2024/1689): medical AI = HIGH-RISK class →
  - transparency, human oversight, documentation; duties 2026-2027
- FDA + Health Canada + MHRA: transparency guiding principles
  - for ML-enabled medical devices (2021, extended 2024)
- Your SHAP beeswarm is not decoration - it is regulatory documentation



The Counterpoint: Stop Explaining Black Boxes?

- Rudin 2019, Nature Machine Intelligence - ~10,000 citations
- Claim: post-hoc explanations APPROXIMATE the model - can be unfaithful
- For HIGH-STAKES decisions: build interpretable models instead
    - (sparse scores, small trees, GAMs) - often NO accuracy loss on tables
- Our ASD cohort agrees: LogReg 0.952 vs random forest 0.937 (test AUC)
- Synthesis: try interpretable first; explain black boxes when they win



Lesson 7 Summary

- Ranking → biomarkers: SHAP, permutation, trees, coefficients
- Explanation → trust: interactions, LIME, counterfactuals, PDP/ICE, IG
- NMTF: fusion embeddings - structure, not labels
- vs Lesson 6: selection builds the model; XAI explains it
- Consensus + panel + enrichment check → biomarkers
- XAI is becoming law (GDPR, AI Act, FDA); Rudin: or interpretable
- Exercise (~20 min): add a pure-noise column to the panel -
    - Gini, permutation (last notebook cell)

Method / paper	Family	Scope	Strength	Watch out
SHAP *Lundberg & Lee 2017 - ~35k cites*	ranking	both	additive, publication std	compute cost
Permutation *Breiman 2001 - ~130k cites*	ranking	global	gold standard, test set	correlated features share
Tree importances *CART 1984 ~60k; XGBoost 2016 ~40k*	ranking	global	free: Gini/gain/weight	biases; name the flavor
Coefficients *classical stats - no single paper*	ranking	global	direction + transparent	standardize first
SHAP interactions *Lundberg et al. 2020 (Nat MI)*	explain	both	epistasis map	p x p cost: panels only
LIME *Ribeiro et al. 2016 - ~25k cites*	explain	local	fast case review	unstable, no additivity
Counterfactuals *Wachter et al. 2017 - ~7k cites*	explain	local	clinical what-if recipe	plausibility constraints
PDP / ICE *Friedman 2001 ~40k; ICE 2015 ~2.5k*	explain	global	effect shapes	assumes independence
Integrated grads *Sundararajan et al. 2017 - ~10k*	explain	local	neural-net standard	baseline choice matters
NMTF *Ding 2006 ~2k; Messa 2024 (new)*	fusion	global	shared-factor embeddings	variance, not labels

ranking → biomarkers | explanation → trust | NMTF → fusion embeddings

Task

1. Copy Xi_train and Xi_test and add a random continuous column to each:

```
rng = np.random.default_rng(0)
Xi_train_noisy = Xi_train.copy()
Xi_train_noisy['noise'] = rng.random(len(Xi_train_noisy))
# same for Xi_test_noisy
```

2.Refit a Random Forest with the same settings as rf_imp (n_estimators=500, random_state=42).

3. Rank all 17 features by Gini importance (feature_importances_) and by permutation importance on the TEST set. Where does noise land in each?

4. Explain the disagreement in 2-3 sentences.